Summary in Graph

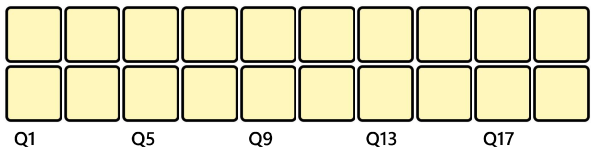
Exam Summary (GO Classes CS Test Series 2026 | Calculus | Topic Wise Test)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	20 (10 + 10)
Total Marks:	30 (10 + 20)
Exam Duration:	60 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (104) out of 1039 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Evaluate the limit

$$\lim_{x \to 0} \frac{1 - \cos x}{\sin^2 x}$$

A. 1
B. $\frac{1}{2}$

C. 2
D. 0

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Determine the value of following limit

$$\lim_{x\to\infty}\left(\sqrt{x^2+4x+1}-x\right)$$

A. 2
B. 4
C. $\frac{1}{2}$
D. 3

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The function $f(x)=x^4-6x^2$ is increasing on the intervals

A. $(0,\sqrt{3})$ only
B. $(\sqrt{3},\infty)$ only
C. $(-\infty,-\sqrt{3})$ and $(0,\sqrt{3})$ only
D. $(-\sqrt{3},0)$ and $(\sqrt{3},\infty)$ only

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The function $f(x)=\cos x-x$

A. is an even function
B. is an odd function
C. is neither an even nor an odd function
D. None of these

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Which of the following functions satisfy the conditions of Rolle's Theorem on the interval $[-1,1]$?

$$f(x)=1-x^{2/3}$$
$$g(x)=x^3-2x^2-x+2$$
$$h(x)=\cos\left(\frac{\pi}{4}(x+1)\right)$$

Rolle's Theorem applies to:

A. both f and g
B. both g and h
C. g only
D. h only

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Suppose that the derivative of a function h is given by:

$$h'(x) = x(x-1)^2(x-2)$$

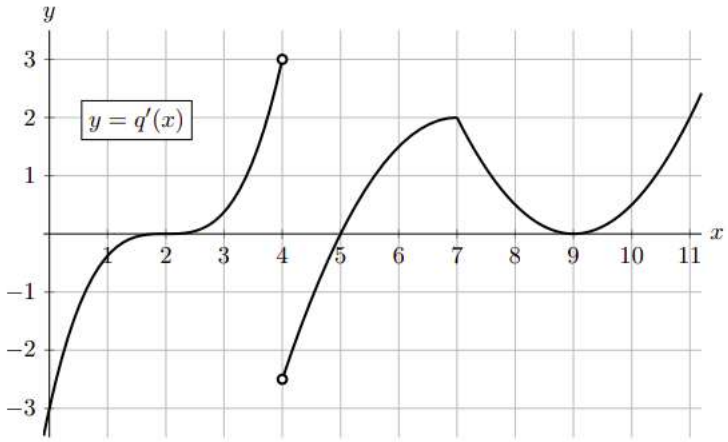
On what interval(s) is h increasing?

A. $(-\infty, 0)$
B. $(-\infty, 0)$ and $(2, \infty)$
C. $(0, 2)$
D. $(0, 1)$ and $(2, \infty)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Calculus

Let $q(x)$ be a continuous function which is defined for all real numbers. A portion of the graph of $q'(x)$, the derivative of $q(x)$, is shown below.



On which of the following interval(s) is $q(x)$ increasing?

A. $(0, 2)$
B. $(2, 4)$
C. $(7, 9)$
D. None of these

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Calculus

Choose the CORRECT statement -

A. The function $f(x) = \exp\left(-x^2\right) - 1$ has the root $x = 0$.
B. If a function f is differentiable on $[-1, 1]$, then there is a point x in that interval where $f'(x) = 0$.
C. If 1 is a root of f, then $f'(x)$ changes sign at 1.
D. If $f''(0) < 0$ and $f''(1) > 0$ then there is a point in $(0, 1)$, where f has an inflection point.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 02sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Evaluate $y''(1)$ where $y = e^x + x^e$.

A. 0
B. 1
C. e^2
D. e

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Consider the following statements:

I. $f(x)$ is continuous on $[a, b]$
II. $f(x)$ is differentiable on (a, b)
III. $f(a) = f(b)$

Which of the above statements are required in order to guarantee a $c \in (a, b)$ such that $f'(c)(b - a) = f(b) - f(a)$?

A. I only
B. I and II only
C. I, II, and III
D. I and III only

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Let $I = (a, b)$ be an open interval and let f be a function which is differentiable on I. Which of the followings is/are true statements -

A. If $f'(x) = 0$ for all $x \in I$, then there is a constant r such that $f(x) = r$ for all $x \in I$.
B. If $f'(x) > 0$ for all $x \in I$, then $f(x)$ is strictly increasing on I.
C. If $f'(x) < 0$ for all $x \in I$, then $f(x)$ is strictly decreasing on I.
D. If $f'(x) > 0$ for all $x \in I$, then $f(x)$ is strictly decreasing on I.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Which of the following is/are FALSE?

A. The absolute maximum value of $f(x) = \dfrac{1}{x}$ on the interval $[2, 4]$ is 2.
B. If $f(x)$ is a continuous function and $f(3) = 2$ and $f(5) = -1$, then $f(x)$ has a root between 3 and 5.
C. The function $g(x) = 2x^3 - 12x + 5$ has 5 real roots.
D. If $h(x)$ is a continuous function and $h(1) = 4$ and $h(2) = 5$, then $h(x)$ has no roots between 1 and 2.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Suppose $g(x)$ is a polynomial function such that $g(-1) = 4$ and $g(2) = 7$. Then there is a number c between -1 and 2 such that

A. $g(c) = 1$
B. $g'(c) = 1$
C. $g(c) = 0$
D. $g'(c) = 0$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Which of the following limit is/are correct?

A. $\lim\limits_{x\to\infty} \sqrt[x]{x} = 1$
B. $\lim\limits_{x\to\infty} \sqrt[x]{x} = e$
C. $\lim\limits_{x\to\infty} \left(1 + \dfrac{2}{x}\right)^x = e^2$
D. $\lim\limits_{x\to\infty} \left(1 + \dfrac{2}{x}\right)^x = e$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Suppose f is twice differentiable with

$$f''(x) = 7x - 2, \quad f'(-2) = 0, \quad \text{and} \quad f(-2) = -2.$$

Find $f(0)$.

A. $-337/6$
B. $-74/3$
C. $23/9$
D. $37/4$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

The sum of three positive numbers is 12 and two of them are equal. Find the largest possible product.

A. 86
B. 64
C. 48
D. 72

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If $f(x) = e^x g(x), g(0) = 2$ and $g'(0) = 1$, then $f'(0)$ is

A. 1
B. 3
C. 2
D. 0

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let f be differentiable for all x. If $f(1) = -2$ and $f'(x) \geq 2$ for $x \in [1, 6]$, then

A. $f(6) \geq 8$
B. $f(6) < 8$
C. $f(6) < 5$
D. $f(6) = 5$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

The equation $x^5 + x + 1 = 0$ has a solution in the interval

A. $[0, 1]$
B. $[-1, 0]$
C. $[-2, -1]$
D. $[1, 2]$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Which of the following expression evaluates to given integral

$$\int \frac{\ln(\ln x)}{x \ln x} dx$$

A. $\frac{\ln x}{x} + C$

B. $\frac{1}{2}(\ln \ln x)^2 + C$

C. $(\ln x)^2 + C$

D. $(\ln \ln x) + C$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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